BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 220 · A*: heuristic search I

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why are predecessors recorded when a distance improves?

Q2. Can Dijkstra assume negative edge weights are safe?

Q3. What is 2+3+2?

Q4. Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

admissible heuristic — An estimate that never exceeds the true cheapest remaining cost.

New learning

g, h and f have different meanings — g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

Start with a frontier — From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

Improve a discovered route — From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

Stop on the selected goal — When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

Keep the heuristic safe — The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

Worked example

Trace A* on P, Q, R, G using the supplied admissible estimates.

1. Predict the next state — From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.

2. Update the table — Through Q, R improves to g4,h2,f6; G gets g11.

3. Update the table — From R, G improves to g6,h0,f6.

4. Update the table — Select G: predecessor chain yields P→Q→R→G, cost 6.

We Do · Practise with guidance

Trace A* on M, N, O, G using the supplied estimates.

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Edges: M-N=2, M-O=5, N-O=2, N-G=7, O-G=2.
h(M)=4, h(N)=3, h(O)=1, h(G)=0.
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Predict the next state

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

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From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

Update the table

From N, O improves to g4,h1,f5.

From O, G improves to g6,h0,f6.

Select G and reconstruct M→N→O→G, total cost 6.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

A node has g=7 and h=3. Give f and define each value.

Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

Exit Ticket · Answer before checking

A* chooses the next vertex solely by its smallest heuristic h(n).

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. They allow reconstruction of the actual shortest route.

Q2. No. Its shortest-path guarantee requires non-negative weights.

Q3. 7.

Q4. 6.

You Do pages · Independent questions

Q1. [4] f=10 [1]; g is known travel cost so far [1]; h is estimated remaining cost [1]; f is estimated total [1].

Q2. [6] From S, A has g2,h2,f4 [2]; B has g5,h1,f6 [2]. Expanding A improves B to g3,f4 [1]; route S→A→B→G total 5 [1].

Q3. [6] A heuristic is admissible when it never overestimates the true cheapest remaining cost [2]. From A the direct cheapest cost to G is 4 [2], so h(A) may be at most 4 [1] (assuming non-negative estimates); overestimating can lead to the optimal path being missed under early-goal termination [1].

Exit Ticket · Model answer

False. It normally chooses the frontier vertex with smallest f(n)=g(n)+h(n).

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: